Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Transplanting

- **Transplanting requirements:** Select vigorous rainy season tomato (*Solanum lycopersicum*) seedlings that are 25–30 days old, 10–15 cm tall, and have 6–8 leaves; avoid using wilted, diseased, or damaged seedlings. Seedling stem strength, root development, physiological maturity, and uniformity standards are Not specified in the provided sources.
- **Transplanting time and schedule:** Rainy season transplanting is documented for tomato crops. Recommended transplanting dates, time of day, duration after sowing, and specific suitable weather or environmental conditions during transplanting are Not specified in the provided sources.
- **Field readiness:** A crop rotation break of at least 1 year is required between planting tomatoes and other solanaceous crops (such as chillies, brinjals, capsicum, potato, and tobacco), cucurbits, or other vegetables to prevent continuous cropping on the same field. Field selection, soil type, soil texture, drainage, soil moisture prerequisites, land preparation, and field leveling before transplanting are Not specified in the provided sources.
- **Soil preparation and amendments:** Pre-emergence herbicides such as Basalin at 1 kg a.i./ha or Pendimethalin at 1 kg a.i./ha, or plastic mulching combined with Pendimethalin at 0.75 kg a.i./ha or Oxyfluorfen at 0.12 kg a.i./ha, can be applied for weed control. Tillage, ploughing, harrowing, bed preparation, ridges and furrows, raised beds, soil sterilization, organic amendments, and basal fertilizer applications or rates are Not specified in the provided sources.
- **Field layout and planting geometry:** Plant tomato seedlings at a spacing of 30 ($\times$)$90cm.Rowspacingvariations, beddimensions, ridgeandfurrowdimensions, plantingpatterns, an$
- **Transplanting methods:** Plant seedlings manually, ensuring that no pressure is exerted on the seedling stems during planting. Mechanical transplanting procedures, transplanting-hole preparation, planting depth, root-ball handling, root orientation, seedling placement, and soil firming precautions are Not specified in the provided sources.
- **Seedling preparation:** Select vigorous seedlings and reject wilted, diseased, or damaged seedlings. Before planting, dip seedling roots for 10–15 minutes in a solution containing Imidacloprid 17.8 SL at 4 ml and either Metalaxyl-M 31.8 ES

at 6 g or Carbendazim at 10 g per 10 L of water, or immerse roots in a solution of insecticide, fungicide, and bacterial fertilizer. Nursery watering prior to seedling removal and root-ball preparation are Not specified in the provided sources.

- **Root handling and transplant shock:** Root protection is provided by dipping roots in insecticide, fungicide, and bacterial fertilizer solutions before planting, and deep cultivation must be avoided to prevent root damage. Transplant shock, wilting, establishment delays, recovery measures, and factors affecting seedling survival are Not specified in the provided sources.

- **Water management:** Provide a light irrigation 3–4 days after transplanting. Maintain irrigation intervals according to soil type and rainfall, applying water every 7–8 days during kharif, 10–12 days during rabi, and 5–6 days during summer. Avoid heavy irrigation following a long drought spell to prevent fruit cracking, and avoid water stress during critical stages of flowering and fruit development. Irrigation timing before, during, or immediately after transplanting and exact water quantities are Not specified in the provided sources.

- **Nutrient management:** Immersion of seedling roots in a bacterial fertilizer solution prior to transplanting is documented. Basal fertilizer prescriptions, starter nutrients, organic manure application rates, fertilizer placement, and application timing during transplanting are Not specified in the provided sources.

- **Environmental requirements:** Irrigation frequency must be adjusted for kharif, rabi, and summer seasonal rainfall and weather variations. Temperature, humidity, sunlight, wind conditions, and explicit soil moisture thresholds during transplanting are Not specified in the provided sources.

- **Plant protection and sanitation:** Dip seedling roots for 10–15 minutes prior to transplanting in a solution of Imidacloprid 17.8 SL at 4 ml combined with either Metalaxyl-M 31.8 ES at 6 g or Carbendazim at 10 g per 10 L of water to manage seedling-borne pathogens and insect pests. Soil-borne disease prevention products, active ingredients, concentrations, and soil sterilization treatments during transplanting are Not specified in the provided sources.

- **Gap filling and establishment monitoring:** Not specified in the provided sources.

- **Post-transplant management:** Maintain a weed-free field during initial plant growth through frequent shallow cultivation, two to three hoeings, earthing up, black or transparent plastic mulching, or pre-emergence herbicides (Basalin at 1 kg a.i./ha or Pendimethalin at 1 kg a.i./ha coupled with one hand weeding 45 days after transplanting). Perform staking 2–3 weeks after transplanting for hybrids and indeterminate varieties using wooden stakes or overhead parallel wires to which individual plants are tied. Shading procedures are Not specified in the provided sources.

- **Transplanting equipment and materials:** Wooden stakes, overhead wires, black or transparent plastic mulching, and root-dip treatment containers are documented for transplanting and field establishment. Specialized machinery, tools, trays, and transport containers are Not specified in the provided sources.

- **Scheduling and recordkeeping:** Transplant rainy season seedlings at 25–30 days old, irrigate lightly 3–4 days after transplanting, stake plants 2–3 weeks after transplanting, and perform hand weeding 45 days after transplanting. Nursery sowing dates, planned transplanting dates, actual transplanting dates, field identification, plant population tracking, and formal survival recordkeeping are Not specified in the provided sources.

- **Effects on crop performance:** Selecting vigorous seedlings prevents disease, avoiding stem pressure during planting prevents mechanical damage, shallow cultivation prevents root injury, avoiding water stress during flowering and fruit development prevents yield loss, avoiding heavy irrigation after drought prevents fruit cracking, and staking facilitates intercultural operations while preserving fruit quality. Quantitative relationships regarding planting depth and root damage are Not specified in the provided sources.

- **Common problems and corrective measures:** Fruit cracking is caused by heavy irrigation following prolonged drought and is corrected by avoiding heavy watering. Root damage and moisture loss are caused by deep cultivation and are corrected by shallow cultivation. Severe weed competition reduces crop yield and is controlled using shallow cultivation, hoeing, earthing up, plastic mulching, and herbicide applications. Transplanting mortality and seedling wilting corrective measures are Not specified in the provided sources.

- **Integrated workflow:** Field establishment begins with site selection adhering to a 1-year solanaceous crop break. Vigorous, disease-free rainy season seedlings (25–30 days old, 10–15 cm tall, with 6–8 leaves) are selected and their roots are dipped for 10–15 minutes in a solution of Imidacloprid 17.8 SL at 4 ml and Metalaxyl-M 31.8 ES at 6 g or Carbendazim at 10 g per 10 L of water. Seedlings are planted at a spacing of 30
(×
)90*cmwithoutexertingpressureonstems.Lightirrigationisapplied*3−−4*daysaftertransplanting*,
−8*daysinkharif*, 10−−12*daysinrabi*, *and*5−−6*daysinsummer*).*Stakingisconducted*2−
−3*weekspost*−*transplant*, *alongsideweedcontrolviashallowcultivation*, *earthingup*, *mulching*, *o*
fillingproceduresareNotspecifiedintheprovidedsources.